

SMARTWATCHES AND WRISTBANDS TO KEEP YOU FIT



cifically with Android OS 4.4 + or iOS 9.3 + . Storage should be 4GB or more, and memory 1GB or more. Operating system can be Wear OS by Google. It must have sensors like accelerometer, altimeter, ambient light, gyroscope, heart rate, microphone, NFC and untethered GPS.

It must have Bluetooth 4.2 Low Energy (BLE) and Wi-Fi 802.11 b/g/n connectivity. Battery life should ideally be 24-plus hours with a rapid charging option and a good processor. Other basic features you may look out for are heartrate tracking, GPS to track commute distance, waterproofing, notifications, personalised dialing, music, interchangeable straps and so on.

A smartwatch uses Bluetooth technology to connect to a phone. A gentle buzz lets the wearer know right away whenever there is a notification, including text, email, social media, app alerts, alarm clock, calendar alerts and the like.

The watch should function well in any temperature with a touchscreen command. Activity-tracking feature enables tracking of steps, distance and calories burned. A customisable dial adds good aesthetics, and custom goal tracking increases usability.

To avoid damage to the watch, only the included charger should be used. The smartwatch should be kept more than twenty centimetres away from implanted medical devices to minimise RF interference.

Fitness band versus smartwatch

A fitness tracker measures only the various aspects of fitness, including heartrate, calories burned, distance covered and sometimes sleep quality. So, for a fitness-focused individual, a

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Wearables are the most convenient tools to collect health data, and monitor and interact with users on-the-go. Smartwatches and activity trackers can remind when a routine task has been missed or left undone. These also promote healthy lifestyles and assist in healthcare wherever and whenever required.

Sensors in smartwatches and wristbands collect data and use AI algorithms to analyse activities in real time. AI algorithms can classify data, such as normal

or irregular heartbeat, and help doctors interpret results in real time or later at convenience. This data is analysed using machine learning (ML).

These devices do not require a set of powerful CPUs and GPUs to carry out ML tasks. The data is analysed and compared using bit arrays, which tremendously reduces memory usage.

ML relies on algorithms, superfast hardware and a huge amount of data for accurate answers. The whole purpose of ML is to allow a software-run machine to recognise what a user is currently doing, while at the same time offer information or an experience that is relevant to the current situation.

Features to look for

A smartwatch should be compatible with Android and iOS smartphones, spe-